

PROJECTILE MOTION

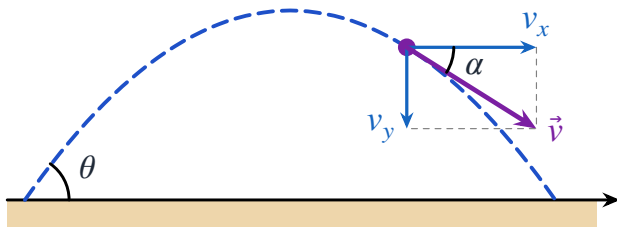
Components • Trajectory • T, H

JEE & NEET
PHYSICS

1 Two Independent Motions

$$u_x = u \cos \theta \text{ (constant)}$$

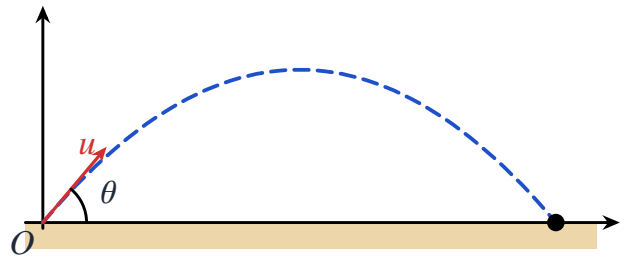
$$v_y = u \sin \theta - gt$$



Horizontal: no acceleration. Vertical: g downward. Time links them.

2 Trajectory Equation

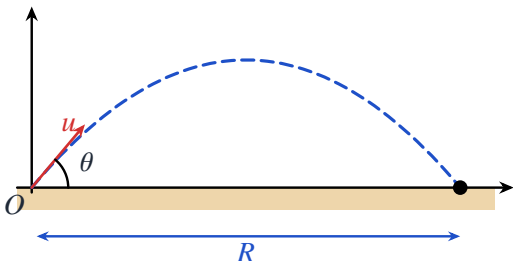
$$y = x \tan \theta - \frac{g x^2}{2u^2 \cos^2 \theta}$$



A parabola. Also $y = x \tan \theta (1 - x/R)$.

3 Time of Flight

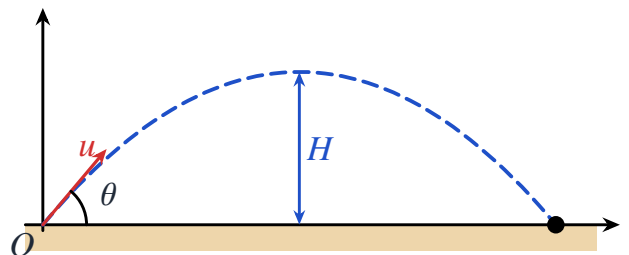
$$T = \frac{2u \sin \theta}{g}$$



Depends only on the vertical part of u :
same u_y , same T .

4 Maximum Height

$$H = \frac{u^2 \sin^2 \theta}{2g}$$



Reached at $t = T/2$, where $v_y = 0$ and
 $v = u \cos \theta$.

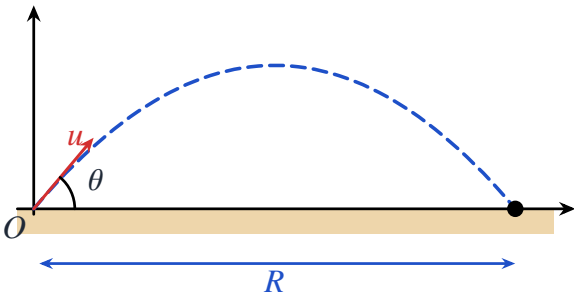
PROJECTILE MOTION

Range • Angles

JEE & NEET
PHYSICS

5 Horizontal Range

$$R = \frac{u^2 \sin 2\theta}{g} = u_x T$$



Same launch and landing level.

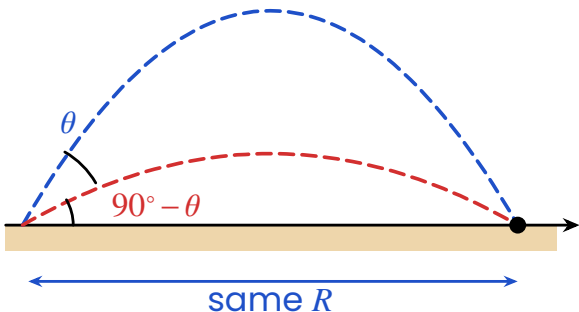
6 Maximum Range

$$R_{\max} = \frac{u^2}{g} \text{ at } \theta = 45^\circ$$
$$H = \frac{R_{\max}}{4} \text{ then}$$

$\sin 2\theta$ is largest at $2\theta = 90^\circ$.

7 Complementary Angles

$$R(\theta) = R(90^\circ - \theta)$$



Same range; $T_1 T_2 = 2R/g$ and
 $H_1 + H_2 = u^2/2g$.

8 Range and Height

$$R = 4H \cot \theta, \quad \tan \theta = \frac{4H}{R}$$

If $R = 4H$, then $\theta = 45^\circ$.

PROJECTILE MOTION

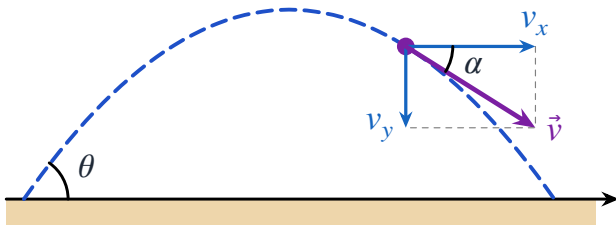
Velocity • Curvature

JEE & NEET
PHYSICS

9 Velocity at Time t

$$v = \sqrt{u^2 - 2ugt \sin \theta + g^2 t^2}$$

$$\tan \alpha = \frac{u \sin \theta - gt}{u \cos \theta}$$

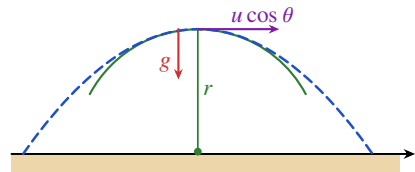


Always along the tangent to the path.

10 Radius of Curvature

$$r = \frac{v^2}{g \cos \alpha}$$

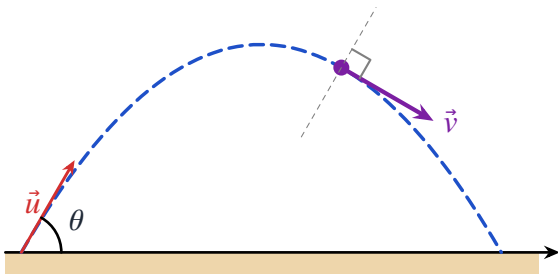
$$r_{\text{top}} = \frac{u^2 \cos^2 \theta}{g}$$



Only the part of g normal to $\vec{v}$ bends the path.

11 Velocity Perpendicular to u

$$t = \frac{u}{g \sin \theta}$$



Possible only if $\theta \geq 45^\circ$ (before landing).

12 Quick Numerical

$u = 20 \text{ m/s}$ at 30° , $g = 10 \text{ m/s}^2$. Find T , H , R .

$$T = \frac{2(20)(\frac{1}{2})}{10} = 2 \text{ s}$$

$$H = \frac{(10)^2}{20} = 5 \text{ m}$$

$$R = \frac{400 \sin 60^\circ}{10} = 20\sqrt{3} \approx 34.6 \text{ m}$$

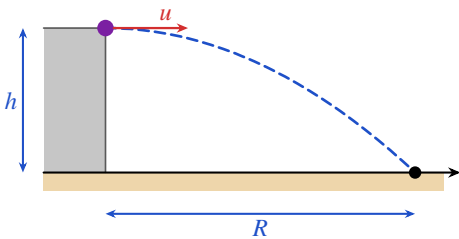
PROJECTILE MOTION

From a Height • Incline

JEE & NEET
PHYSICS

13 Thrown Horizontally

$$T = \sqrt{\frac{2h}{g}}, \quad R = u\sqrt{\frac{2h}{g}}$$

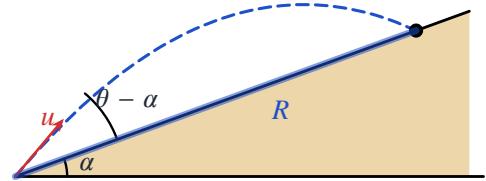


Lands with $v = \sqrt{u^2 + 2gh}$; $\tan \beta = gT/u$ below horizontal.

14 On an Incline (up)

$$T = \frac{2u \sin(\theta - \alpha)}{g \cos \alpha}$$

$$R = \frac{2u^2 \sin(\theta - \alpha) \cos \theta}{g \cos^2 \alpha}$$



θ from the horizontal, R along the incline.

15 Best Angle on an Incline

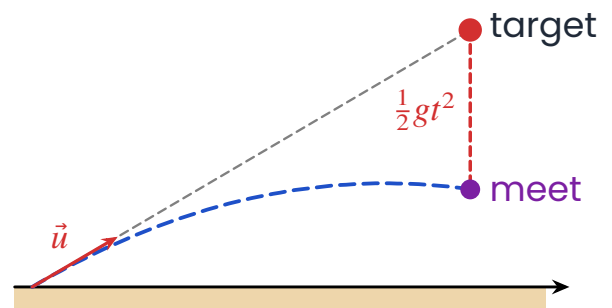
$$\theta = 45^\circ + \frac{\alpha}{2}$$

$$R_{\max} = \frac{u^2}{g(1 + \sin \alpha)}$$

Down the incline: $\theta = 45^\circ - \alpha/2$,
 $R_{\max} = u^2/g(1 - \sin \alpha)$.

16 Monkey and Hunter

$$\vec{a}_{\text{rel}} = \vec{g} - \vec{g} = 0$$



Aim straight at a falling target and you always hit it (if it is in range).

PROJECTILE MOTION

Quick Revision • Traps

JEE & NEET
PHYSICS

17 Everything at a Glance

Quantity	Formula	Note
T	$2u \sin \theta / g$	only u_y matters
H	$u^2 \sin^2 \theta / 2g$	at $t = T/2$
R	$u^2 \sin 2\theta / g$	max u^2 / g at 45°
Path	$y = x \tan \theta (1 - x/R)$	a parabola
Top	$v = u \cos \theta$	$r = u^2 \cos^2 \theta / g$
From h	$T = \sqrt{2h/g}$	$R = uT$

18 Common Traps

- At the top $v \neq 0$: it is $u \cos \theta$.
- g never changes; only v_y changes.
- Equal T means equal u_y , not equal u .
- R formula needs launch and landing at one level.

19 Two Projectiles

$$\vec{r}_{12} = (\vec{u}_1 - \vec{u}_2) t$$

Both fall with g , so one sees the other move in a straight line at constant velocity.